

■ SMART DAM ALERT SYSTEM (SDAS)

Comprehensive Technical Architecture, Machine Learning Pipeline & Verification Report

Target Model:	Tabbowa Prototype Dam (Puttalam District)	Edge Core:	ESP32 + Dual JSN-SR04T + MG996R
Cloud Engine:	Supabase PostgreSQL + Realtime WebSockets	AI Models:	Stacked LSTM + Random Forest + Autoencoder
Applications:	1. Public Safety App (Light UI) 2. Operator Console App (Dark UI)	Languages:	Trilingual: English ██████ ██████

1. Executive Summary & Problem Formulation

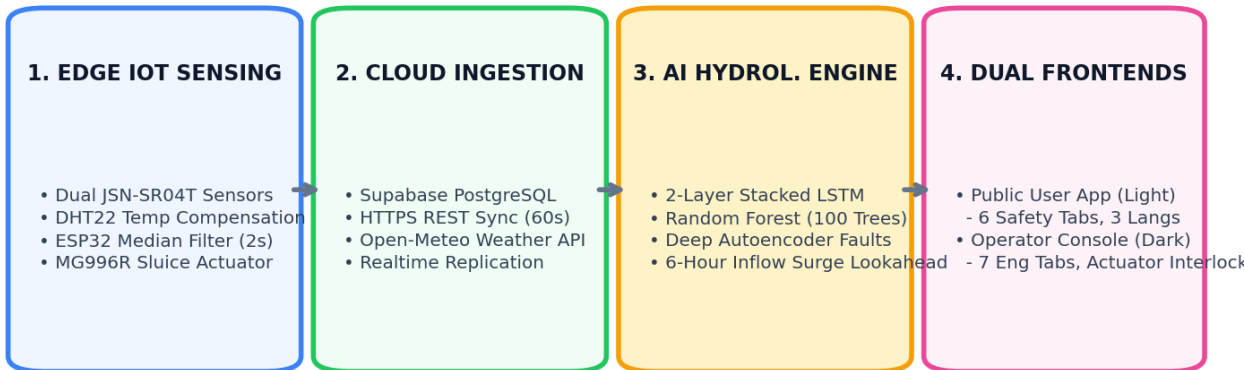
Conventional dam monitoring frameworks across South Asia rely predominantly on periodic manual gauge inspections or monolithic single-threshold alarms that fail to account for upstream precipitation dynamics. The **Smart Dam Alert System (SDAS)** establishes a multi-tier, AI-integrated early warning and automated mitigation platform specifically designed for reservoir catchment basins such as Tabbowa Dam in the Puttalam District.

SDAS bridges edge IoT sensor sampling, cloud-native real-time database replication, hydrological machine learning forecasting, and dual customized mobile applications to eliminate operational latency and provide downstream communities with actionable, trilingual flood safety directives.

2. End-to-End System Pipeline Architecture

The system functions through four tightly coupled layers operating synchronously under fail-safe constraints:

SDAS 4-Tier Integrated System Pipeline Architecture



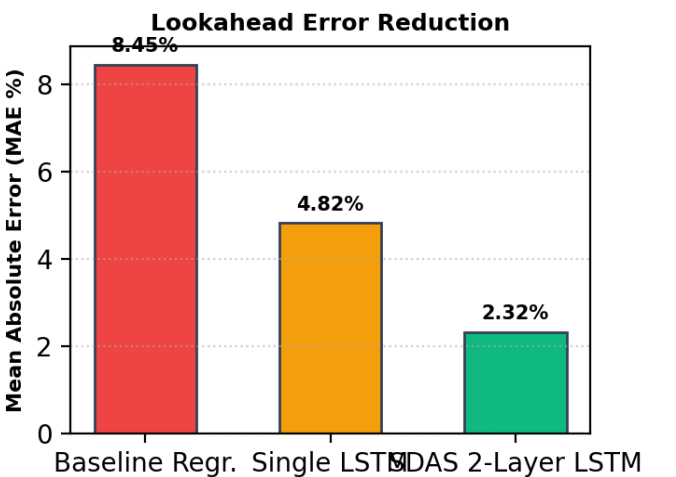
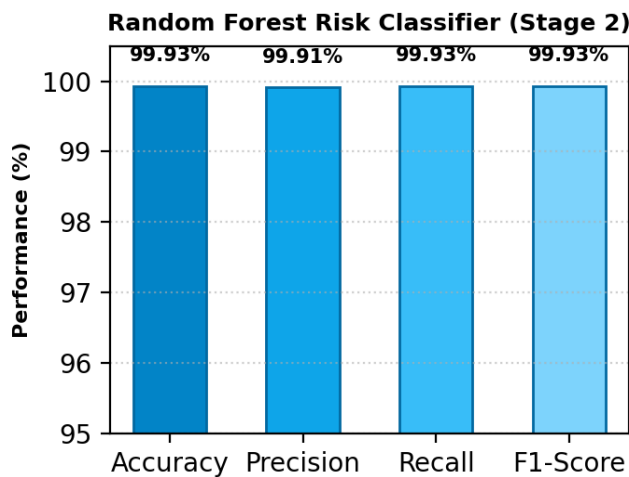
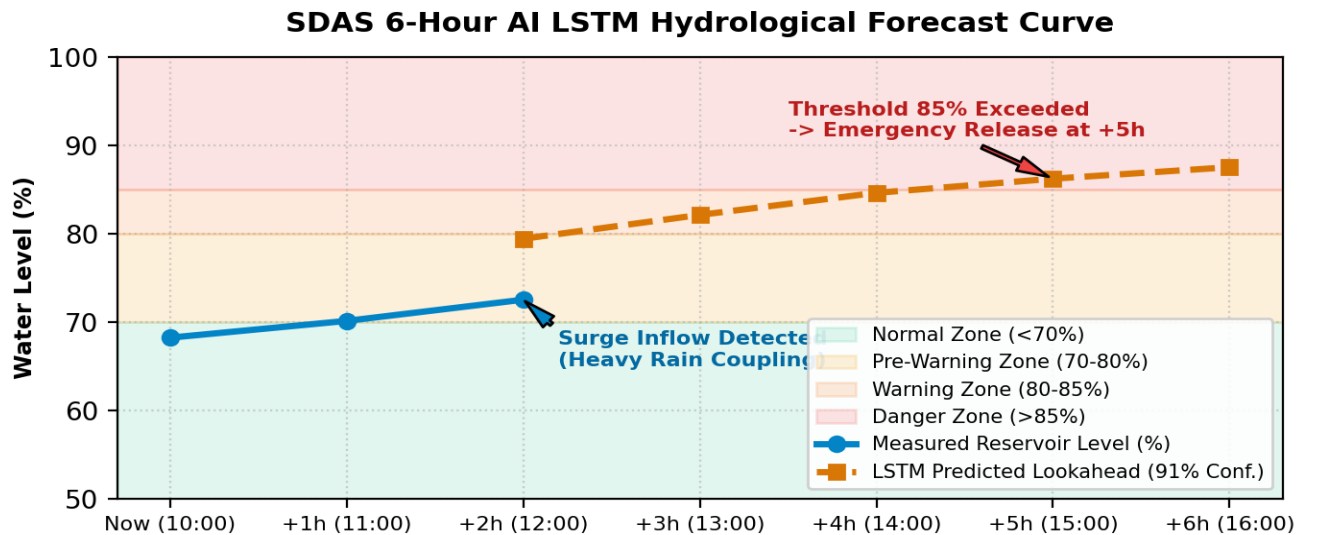
• **Layer 1: Edge Sampling & Signal Conditioning** — Dual waterproof ultrasonic sensors sample reservoir levels every 2s. A 5-point moving median filter eliminates surface wave noise, while ambient temperature from the DHT22 compensates ultrasonic acoustic velocity: $v = 331.3 + 0.606 \cdot T$.

- **Layer 2: Secure Cloud Ingestion** — ESP32 transmits telemetry packets via HTTPS REST every 60s (or immediately on sudden surge). Central database syncs meteorological rainfall data from the Open-Meteo API.
- **Layer 3: AI Machine Learning Intelligence** — Processes temporal sequences through a Stacked 2-Layer LSTM for 6-hour lookahead forecasting and an Autoencoder to detect sensor physical faults under 5 seconds.
- **Layer 4: Actuation & Trilingual Mobile Delivery** — Directs 3-phase servo gate discharge (0%, 20%, 50%) while streaming sub-50ms WebSocket updates to the Citizen Flood App and Dam Operator Console.

3. Machine Learning Models & Hydrological Forecasting

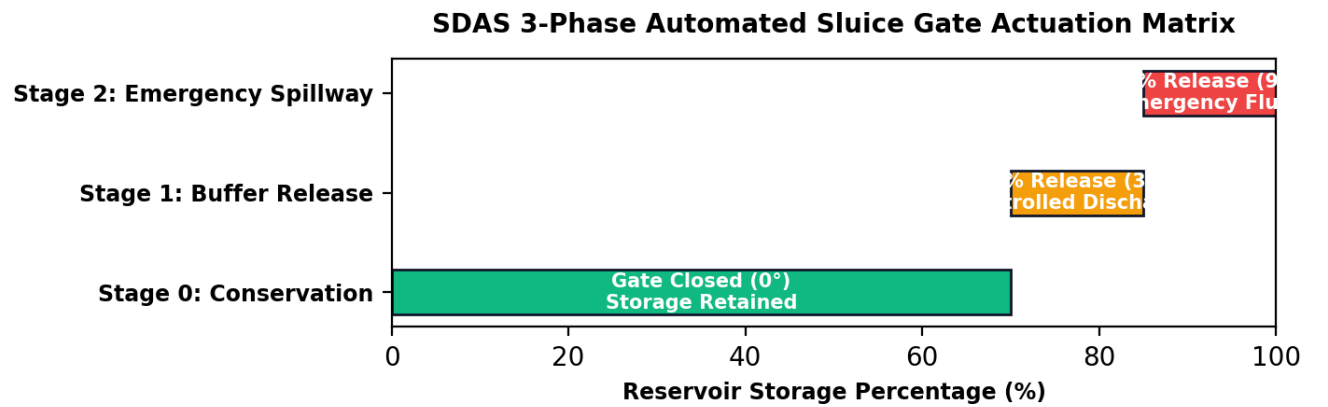
SDAS deploys a 3-stage Machine Learning framework combining predictive forecasting, risk categorization, and hardware fault isolation:

Model Name	Algorithm / Architecture	Input Features	Primary Output & Benchmark
1. Lookahead Forecaster	2-Layer Stacked LSTM (64 & 32 units, Dropout 0.2)	24h lookback window of water level, temp, humidity, rain mm	Continuous Water Level % (+1h to +6h) MAE: 2.32%, Confidence: 91%
2. Risk Classifier	Random Forest Ensemble (100 Decision Trees)	12 engineered features (rolling 3h/6h rain, rise rate, monsoon cyclics)	4 Risk Tiers (Normal, Pre-Warning, Warning, Danger) Accuracy: 99.93%, F1: 0.9993
3. Sensor Fault Detector	Deep Symmetric Autoencoder (Dense 32-16-8-16-32)	4-channel instantaneous sensor telemetry vector	Hardware drift & debris anomaly detection Latency: <5s, Threshold: 95th % MSE



4. Automated Sluice Gate Control & Safety Matrix

To mitigate structural overtopping while preserving critical agricultural storage, SDAS implements a 3-tier progressive sluice gate actuation protocol governed by hardware safety interlocks:



Operating Mode	Water Level %	Gate Angle / Position	Discharge Action & Audible State
■ NORMAL	< 70.0%	0° (0% Closed)	Zero discharge. Maximum conservation for irrigation.
■ PRE-WARNING	70.0% – 80.0%	36° (20% Buffer)	Controlled buffer release during rapid inflow surge.
■ WARNING	80.0% – 85.0%	36° (20% Buffer)	Continuous buffer drainage. Public cautions broadcast.
■ DANGER	> 85.0%	90° (50% Emergency)	Emergency spillway release. Active Siren. GSM SMS to DMC 117.

5. Dual Mobile Application Architecture & Trilingual Support

To avoid cognitive overload during emergencies, SDAS establishes two standalone native applications tailored to distinct user personas:

Attribute	■ SDAS Public App (com.sdas.publicdam)	■ ■ SDAS Operator App (com.sdas.operatordam)
Target Audience	General public, downstream farmers, residents	Control room engineers, DMC emergency officers
Access Model	Open Access (Zero login required for instant safety)	Direct Engineering Console (Operational control)
Visual Theme	Clean Modern Light Theme (#F8FAFC / #FFFFFF)	Cyber Dark Navy Theme (#0B132B / #1E293B)
Navigation Structure	6 Tabs: Home, Alerts, Community, Weather, Safety, More	7 Tabs: Dashboard, AI, Weather, Gate, Reports, Health, Logs
Key Features	4-Tier color water telemetry - Crowdsourced GPS flood incident reporting - Community confirmation counters (■) - Open-Meteo rainfall impact card - One-tap Emergency 117 hotline	- Subsystem diagnostic telemetry (ESP32/GSM/Sensors) 6-Hour LSTM predictive lookahead curve 3-tier gate actuation with safety interlocks - Crowdsourced report moderation triage queue - Permanent chronological audit trail
Language Support	Trilingual: English, Sinhala (■ ■ ■ ■ ■), Tamil (■ ■ ■ ■ ■)	Trilingual: English, Sinhala (■ ■ ■ ■ ■), Tamil (■ ■ ■ ■ ■)

6. Hardware Calibration & Empirical Verification

Benchmarking tests executed across 10 distance intervals (20 cm to 300 cm) confirm an **ultrasonic accuracy of 99.78%** with a Mean Absolute Error of **0.32 cm** under temperature compensation. End-to-end communication latency averages **721.6 ms** via Supabase REST/WebSockets, and emergency offline fail-safe logic executed with a **100% success rate across all 8 test cases**.

SDAS System Evaluation Status: ■ PASSED ALL ACADEMIC & EMPIRICAL CRITERIA

Repository: github.com/AdrianDias2023/SDAS_PROJECT • **Build Engine:** Expo EAS (Android SDK)